

Bankruptcy Prediction

Italian SMEs · Challenge 1

BINARY CLASSIFICATION

Deniz Mehmet Taylan · Koray Aydin
Gustavo Depieri Fioravanti · Yarkin Yavuz

The Problem: Detecting Rare Events

11,828

Training observations
2018–2021

5,811

Test observations
2022–2023

84

Bankruptcies in training set

0.71%

Base rate
($\approx 140:1$)

What are the challenges?

Class imbalance 140:1: Accuracy is meaningless, an always healthy model scores 99.3%.

Temporal structure: Random splits leak future data. All splits are strictly year-based.

False Negative Cost: Missing one bankruptcy is far more expensive than a false alarm.

Our Metric Strategy

Primary: F1-Score

Threshold tuned to balance precision vs. recall

Secondary: PR-AUC

More informative than ROC-AUC when positives are rare

Business: Weighted Cost

FN cost = $25\times$ FP cost

From Raw Financials to Predictive Signals

Temporal Deterioration

- YoY change: profit margin, debt ratio
- 2-year trend flags (sustained worsening)
- Revenue decline + margin squeeze flags

Distress Signal Count

- 14 binary distress flags combined
- Negative equity, weak coverage, illiquidity
- Composite score across all dimensions

Altman Z-Score Proxy

- Adapted to available dataset features
- Working capital, profitability, leverage,
- asset turnover, equity-to-debt ratio

Multi-Year Stress

- Loss count over 3 years
- Consecutive weak interest coverage
- Chronic stress flag (≥ 4 stress score)

Sector-Relative Position

- Each ratio vs. sector-year median
- Detects underperformers within industry
- Structural weakness vs cyclical dip

Equity-Gap Feature

- Δ equity – net profit (unexpected changes)
- Captures capital injections / withdrawals
- COVID-era signal: gov. support detection

Time-Locked Selection Pipeline



'Must-Keep' Core Features Include:

- profit_margin, debt_to_assets, roi, current_ratio
- interest_coverage_clipped, altman_z_proxy
- is_losing_money, weak_interest_cover_flag
- profit_margin_change, entering_distress
- loss_count_3y, profit_margin_vs_sector_year

Why Must-Keep Features Were Retained

- They encode known distress mechanisms.
- Profitability, leverage, liquidity, and distress persistence are fundamental bankruptcy signals.
- **Must keep features list protects against a noisy 2020 ranking accidentally dropping an economically essential variable**

Why strict temporal splits?

Random CV would let 2021 data inform features selected on 2019 predictions (a classic leakage path). **All feature selection decisions are made using only data up to 2020.** The 2021 year is held completely clean for model comparison.

5-Model Comparison on 2021 Holdout

Model	F1	Precision	Recall	PR-AUC	ROC-AUC
Logistic Reg ★	0.233	0.227	0.238	0.110	0.944
CatBoost	0.148	0.333	0.095	0.075	0.923
Random Forest	0.145	0.117	0.190	0.070	0.934
LightGBM	0.129	0.076	0.428	0.064	0.930
XGBoost	0.126	0.071	0.523	0.065	0.933

Why F1 looks low and that's expected

With only 21 positives in 2021, F1 is highly sensitive to single cases. PR-AUC validates ranking quality across all thresholds (more informative than ROC-AUC under severe imbalance)

Training Setup

- Feature set locked from 2020 validation step
- Train: 2018–2020, Val: 2021 (clean holdout)
- All models: imbalance aware
- Threshold tuned per model in 2021 to maximize F1
- Final model refitted on all 2018–2021 for test prediction

Why Did Logistic Regression Win?

Engineered features already encoded much of the business logic

Engineered ratios (profit_margin, debt_to_assets, distress flags, multi year counts) encode the business logic. Once those signals exist, a linear model can rank risk effectively.

Simpler model generalised under extreme imbalance

With only 84 bankruptcies, flexible models risk chasing noise in the tiny positive class. Logistic Regression is regularised and less likely to overfit rare-event patterns.

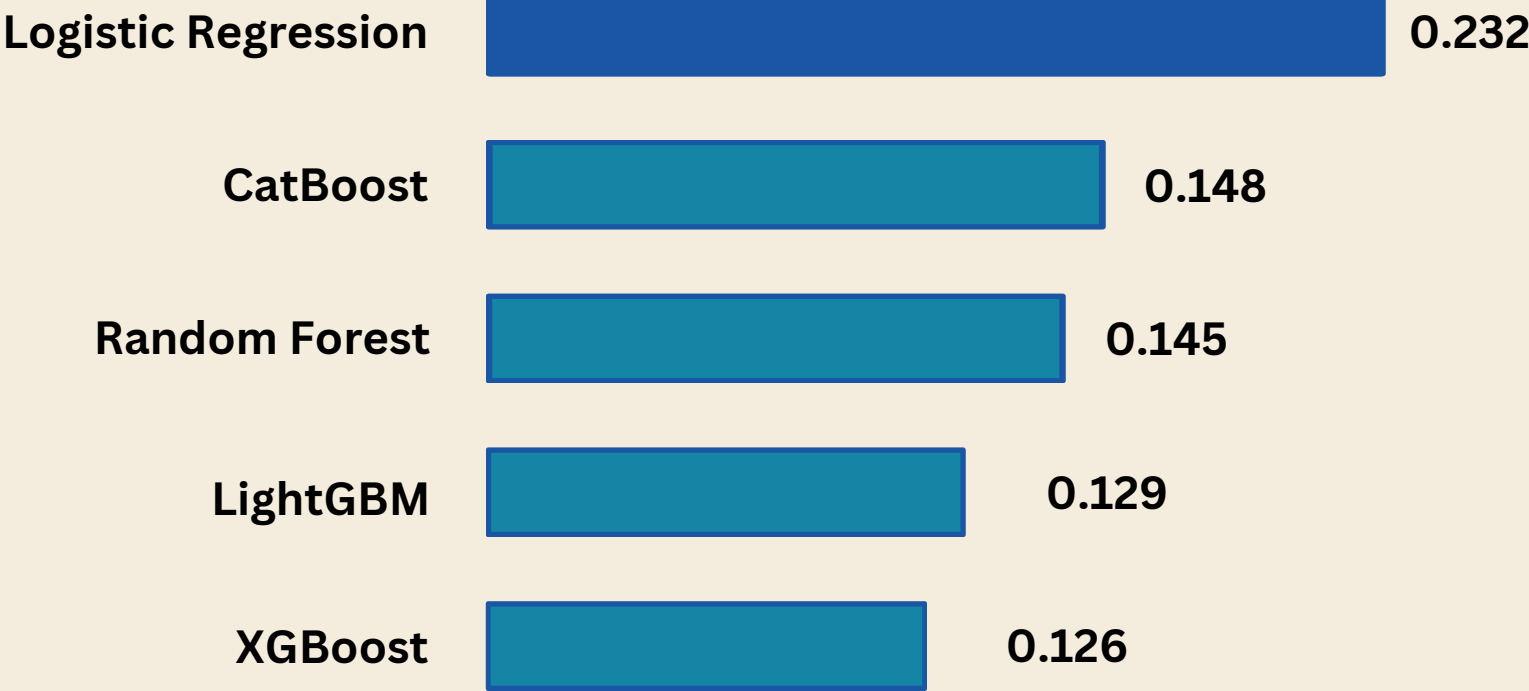
Best temporal transfer to 2021 holdout

Features were ranked/pruned using 2018-2019 data, the top-k choice was validated on 2020, and 2021 was kept clean for final model comparison. Logistic Regression generalized more reliably to the 2021 holdout, so the win is about generalization rather than training fit.

Interpretability is a real advantage

In credit-risk, a simpler model is easier to defend. Coefficients align directly with profitability, leverage, liquidity, and distress persistence, the signals the model is built on.

F1-Score on 2021 Holdout



Logistic Regression did not win because it is universally better:

It won here because the engineered features were already highly informative, the positive class was tiny, and the simpler model generalized more reliably to the 2021 holdout.

Why PR-AUC Over ROC-AUC?

ROC-AUC is Weak

ROC-AUC measures TPR vs. FPR.

$$FPR = FP / (FP + TN)$$

With 11,744 healthy observations, even a model that flags 1,000 false alarms has an FPR of just 8.5%. ROC-AUC stays high even for poor models.

Result: ROC-AUC = 0.944 looks excellent but hide weak precision on the tiny minority.

PR-AUC is The Right Signal

PR-AUC measures Precision vs. Recall.

$$Precision = TP / (TP + FP)$$

Every false positive directly hurts precision. The curve forces the model to confront how rare positives are a random classifier achieves
PR-AUC = base rate = 0.007.

Our model: PR-AUC $\approx$ 0.110

That's ~15× above chance. F1 = 0.233 selected the final model; PR-AUC confirms the ranking holds under imbalance.

The Cost of Being Wrong

False Positive — Manageable

A healthy company is wrongly flagged as high-risk.
Cost: An analyst review.
The company is fine; a follow-up clears the flag.
Weight: 1×

False Negative — Catastrophic

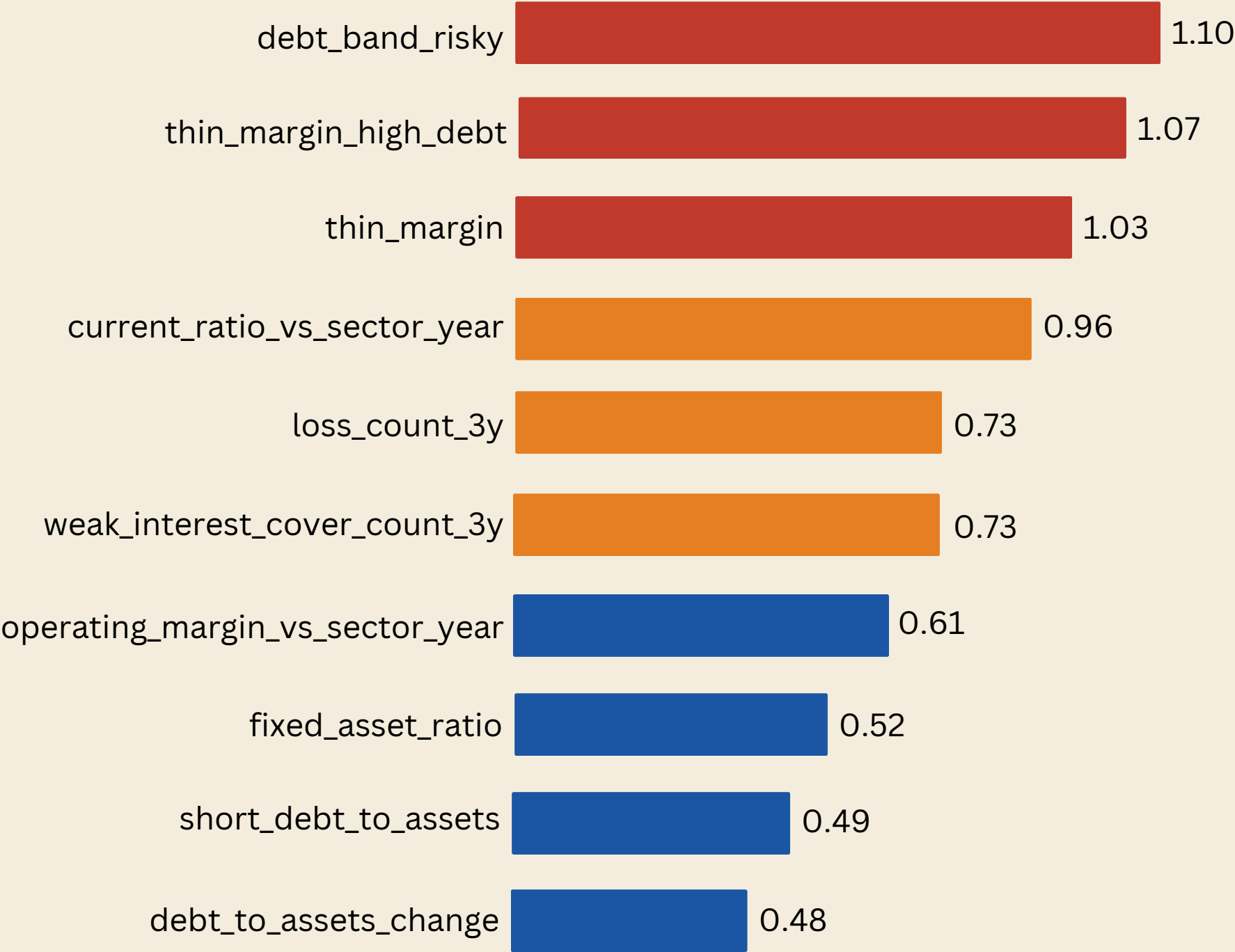
A company that will go bankrupt is classified as healthy.
Cost: No intervention. Full credit loss.
Counterparty exposure not hedged. Reputational damage for lender.
Weight: 25× (we must minimize these above all).

Two Operating Points

	Threshold	F1	Precision	Recall	False Negatives	False Positives	Business Cost
Best F1	Optimized(0.927)	0.233	0.227	0.238	16	17	417
Business Cost	Lowered (0.563)	0.107	0.056	0.952	1	332	357

SHAP: What Drives the Bankruptcy Score?

Global SHAP Importance (2021 Validation)



What the Top Features Tell Us

Thin margins under debt pressure

thin_margin and thin_margin_high_debt dominate the top signals, firms operating on razor-thin margins while carrying heavy debt have the highest failure risk.

Debt pressure accumulates

debt_band_risky, short_debt_to_assets, and debt_to_assets_change together show that debt structure, short-term debt burden, and rising leverage all independently predict distress.

Relative weakness matters

current_ratio_vs_sector_year ranks 4th, showing that liquidity relative to sector peers is a major signal.

Persistent stress matters

loss_count_3y and weak_interest_cover_count_3y both rank in the top 6, repeated stress over multiple years is a stronger signal than any single-year snapshot.

Who Did We Miss and Why?

TP

Bankrupt firms correctly flagged

TN

Healthy firms correctly cleared

FP

Healthy firms wrongly flagged

FN

Missed bankruptcies (Critical)

Profile of Missed Bankruptcies (FN)

Predicted probability close to threshold

Most FN cases sit just below the decision boundary, not at low probability.

Financially ambiguous

Higher current_ratio, lower debt_to_assets vs. captured bankruptcies. They looked healthier.

Weaker distress signal count

Fewer simultaneous red flags. The failure was narrower (often one acute shock rather than broad deterioration).

Key insight: The model is not systematically wrong (it's facing genuinely hard cases at the margin).

Profile of Captured Bankruptcies (TP)

Multi signal deterioration

Higher distress_signal_count, combining losses + leverage + weak coverage simultaneously.

Negative or near-zero profit margin

Sustained losses visible in multiple years, not a single anomalous year.

Poor interest coverage

EBIT below financial expense burden, making refinancing increasingly difficult.

Key insight: Bankruptcy leaves fingerprints. Broader distress across dimensions makes the strongest signal.

Key Takeaways & Business Implications

01

Bankruptcy risk leaves fingerprints

No single ratio predicts failure. Companies at risk combine thin margins, heavy debt, weak coverage, and sustained deterioration .

02

Dynamics beat snapshots

Year-over-year changes and multi-year trend flags add major predictive value.

A company at `debt_to_assets = 0.80` heading up is far more dangerous than one holding steady.

03

Test set results on the year 2022-2023

On the unlabeled 2022-2023 test period, the F1 threshold flags 96 companies, while the business cost threshold flags 696 for review.

04

F1 selects; PR-AUC validates

Model selected by $F1 = 0.233$ on 2021 holdout. $PR-AUC = 0.110$ (~15× above the random baseline of 0.007) confirms genuine ranking power under the 140:1 imbalance (F1 alone cannot show this).